

Math 6601 Homework 4

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Problem 1.

Let $v = (v_1, \dots, v_n^T) \in \mathbf{R}^n$, and suppose that $(v, x) = 0$ for all $x \in \mathbf{R}^n$. Then $v = 0$.

Proof. Let $e_i = (0, \dots, 0, 1, 0, \dots, 0)^T \in \mathbf{R}^n$, so that the i th entry of e_i is 1 and all others are 0 for $i = 1, 2, \dots, n$. Then

$$v_i = (v, e_i) = 0$$

for $i = 1, 2, \dots, n$, so $v = (0, 0, \dots, 0)^T = 0$. □

Problem 2.

Let $A \in \mathbf{R}^{n \times n}$ be symmetric and positive definite, and let $V = \{v^{(1)}, \dots, v^{(n)}\} \subseteq \mathbf{R}^n$ be a set of nonzero, A -orthogonal vectors.

(a) V is linearly independent.

Proof. Let $c_1, \dots, c_n \in \mathbf{R}$, and suppose that

$$c_1 v^{(1)} + \dots + c_n v^{(n)} = 0.$$

Taking the inner product of both sides with $Av^{(i)}$ gives

$$c_i (v^{(i)}, Av^{(i)}) = (c_1 v^{(1)} + \dots + c_n v^{(n)}, Av^{(i)}) = (0, Av^{(i)}) = 0$$

because $(v^{(j)}, Av^{(i)}) = 0$ if $i \neq j$ by the A -orthogonality of V . Furthermore, since A is positive definite, $(v^{(i)}, Av^{(i)}) \neq 0$. This implies that $c_i = 0$. Since $1 \leq i \leq n$ was arbitrary, it follows that $c_1 = c_2 = \dots = c_n = 0$. Thus, V is linearly independent. □

(b) If x is orthogonal to every element of V , then $x = 0$.

Proof. Since V is linearly independent and contains n elements, it follows that V is a basis for $\mathbf{R}^n$. Let $v \in \mathbf{R}^n$ be given. Then there exist $v_1, \dots, v_n \in \mathbf{R}$ such that

$$v = v_1 v^{(1)} + \dots + v_n v^{(n)}.$$

Because x is orthogonal to every element of V , we have

$$(x, v) = (x, v_1 v^{(1)} + \dots + v_n v^{(n)}) = v_1 (x, v^{(1)}) + \dots + v_n (x, v^{(n)}) = 0 + \dots + 0 = 0.$$

Apply the result in Problem 1 shows that $x = 0$. □

Problem 3.

Let $\{r^{(k)}\}$ be the residual vectors for a conjugate direction method to solve $Ax = b$, where $x, b \in \mathbf{R}^n$, and $A \in \mathbf{R}^{n \times n}$ is symmetric positive definite, and the direction vectors are $\{v^{(k)}\}$. Then for $k = 1, \dots, n$,

$$(r^{(k)}, v^{(j)}) = 0 \quad \text{if } j \leq k.$$

Proof. First, we show that $(r^{(1)}, v^{(1)}) = 0$. Indeed, since $r^{(1)} = r^{(0)} - t_1 A v^{(1)}$, and $v^{(1)} = r^{(0)}$, we have

$$(r^{(1)}, v^{(1)}) = (r^{(0)} - t_1 A v^{(1)}, v^{(1)}) = (r^{(0)}, r^{(0)}) - t_1 (A v^{(1)}, v^{(1)}).$$

Recalling that $t_1 = \frac{(r^{(0)}, r^{(0)})}{(A v^{(1)}, v^{(1)})}$, we get $(r^{(1)}, v^{(1)}) = 0$.

Second, suppose for the sake of induction that, for some $1 \leq \ell < n$, we have $(r^{(\ell)}, v^{(j)}) = 0$ for $j \leq \ell$. Then, since $r^{(\ell+1)} = r^{(\ell)} - t_{\ell+1} A v^{(\ell+1)}$, we get, for $j \leq \ell$,

$$(r^{(\ell+1)}, v^{(j)}) = (r^{(\ell)} - t_{\ell+1} A v^{(\ell+1)}, v^{(j)}) = (r^{(\ell)}, v^{(j)}) - t_{\ell+1} (A v^{(\ell+1)}, v^{(j)}) = 0;$$

the first term is zero by the induction hypothesis, and the second term is 0 because the direction vectors are A -orthogonal in a conjugate direction method.

Finally, when $j = \ell + 1$, recalling that $t_{\ell+1} = \frac{(v^{(\ell+1)}, r^{(\ell)})}{(A v^{(\ell+1)}, v^{(\ell+1)})}$, we have

$$(r^{(\ell+1)}, v^{(\ell+1)}) = (r^{(\ell)}, v^{(\ell+1)}) - \frac{(v^{(\ell+1)}, r^{(\ell)})}{(A v^{(\ell+1)}, v^{(\ell+1)})} (A v^{(\ell+1)}, v^{(\ell+1)}) = 0.$$

Thus, the result follows for $k = 1, \dots, n$ by induction. □

Problem 4.

Let

$$T_k(t) = \begin{cases} \cos(k \cos^{-1}(t)) & |t| < 1 \\ [\operatorname{sgn}(t)]^k \cosh(k \cosh^{-1}(|t|)) & |t| \geq 1, \end{cases}$$

where sgn is the signum function. Then T_k is a polynomial of degree k for all $k \geq 0$.

Proof. Suppose that $|t| < 1$. Then $T_0(t) = \cos(0) = 1$, and $T_1(t) = \cos(\cos^{-1}(t)) = t$. For $k \geq 1$, we have

$$\begin{aligned} T_{k+1}(t) &= \cos((k+1) \cos^{-1}(t)) \\ &= \cos(k \cos^{-1}(t)) \cos(\cos^{-1}(t)) - \sin(k \cos^{-1}(t)) \sin(\cos^{-1}(t)) \\ &= t T_k(t) - [\sin((k-1) \cos^{-1}(t)) \cos(\cos^{-1}(t)) + \cos((k-1) \cos^{-1}(t)) \sin(\cos^{-1}(t))] \sin(\cos^{-1}(t)) \\ &= t T_k(t) - t \sin((k-1) \cos^{-1}(t)) \sin(\cos^{-1}(t)) - T_{k-1}(t) \sin^2(\cos^{-1}(t)). \end{aligned}$$

Observing that

$$\sin((k-1) \cos^{-1}(t)) \sin(\cos^{-1}(t)) = \cos((k-1) \cos^{-1}(t)) \cos(\cos^{-1}(t)) - \cos(k \cos^{-1}(t)) = t T_{k-1}(t) - T_k(t),$$

we can substitute into the above to get

$$\begin{aligned} T_{k+1}(t) &= tT_k(t) - t(tT_{k-1}(t) - T_k(t)) - T_{k-1}(t)(1 - \cos^2(\cos^{-1}(t))) \\ &= 2tT_k(t) - t^2T_{k-1}(t) - T_{k-1}(t)(1 - t^2) = 2tT_k(t) - T_{k-1}(t). \end{aligned}$$

Similarly, if $|t| \geq 1$, then $T_0(t) = \cosh(0) = 1$, and $T_1(t) = \operatorname{sgn}(t) \cosh(\cosh^{-1}(|t|)) = \operatorname{sgn}(t)|t| = t$. For $k \geq 1$, we have

$$\begin{aligned} T_{k+1}(t) &= [\operatorname{sgn}(t)]^{k+1} \cosh((k+1) \cosh^{-1}(|t|)) \\ &= [\operatorname{sgn}(t)]^{k+1} [\cosh(k \cosh^{-1}(|t|)) |t| + \sinh(k \cosh^{-1}(|t|)) \sinh(\cosh^{-1}(|t|))] \\ &= tT_k(t) + [\operatorname{sgn}(t)]^{k+1} [\sinh(k \cosh^{-1}(|t|)) \sinh(\cosh^{-1}(|t|))]. \end{aligned}$$

Note that

$$\begin{aligned} \sinh(k \cosh^{-1}(|t|)) &= \sinh((k-1) \cosh^{-1}(|t|)) \cosh(\cosh^{-1}(|t|)) + \cosh((k-1) \cosh^{-1}(|t|)) \sinh(\cosh^{-1}(|t|)) \\ &= |t| \sinh((k-1) \cosh^{-1}(|t|)) + [\operatorname{sgn}(t)]^{k-1} T_{k-1}(t) \sinh(\cosh^{-1}(|t|)). \end{aligned}$$

Substituting into the above, we get

$$T_{k+1}(t) = tT_k(t) + t[\operatorname{sgn}(t)]^{k-1} \sinh((k-1) \cosh^{-1}(|t|)) \sinh(\cosh^{-1}(|t|)) + T_{k-1}(t) \sinh^2(\cosh^{-1}(|t|)).$$

Finally, we observe that

$$\begin{aligned} \sinh((k-1) \cosh^{-1}(|t|)) \sinh(\cosh^{-1}(|t|)) &= \cosh(k \cosh^{-1}(|t|)) - \cosh((k-1) \cosh^{-1}(|t|)) \cosh(\cosh^{-1}(|t|)) \\ &= [\operatorname{sgn}(t)]^k T_k(t) - t[\operatorname{sgn}(t)]^k T_{k-1}(t) \\ &= [\operatorname{sgn}(t)]^k (T_k(t) - tT_{k-1}(t)), \end{aligned}$$

which yields

$$\begin{aligned} T_{k+1}(t) &= tT_k(t) + t(T_k(t) - tT_{k-1}(t)) + T_{k-1}(t)(\cosh^2(\cosh^{-1}(|t|)) - 1) \\ &= 2tT_k(t) - t^2T_{k-1}(t) + t^2T_{k-1}(t) - T_{k-1}(t) \\ &= 2tT_k(t) - T_{k-1}(t). \end{aligned}$$

Thus, for all t , the sequence $\{T_k(t)\}$ satisfies the recurrence $T_{k+1}(t) = 2tT_k(t) - T_{k-1}(t)$ for $k \geq 1$ subject to initial conditions $T_0(t) = 1$ and $T_1(t) = t$. It follows from a simple inductive argument that T_k must be a polynomial of degree k for all k . Indeed, we observe that $T_0(t) = 1$, and $T_1(t) = t$, which are polynomials of degree 0 and 1. Suppose for some $k \geq 1$ that T_{k-1} and T_k are polynomials of degree $k-1$ and k . Then

$$T_{k+1}(t) = 2tT_k(t) - T_{k-1}(t)$$

is a polynomial of degree $k+1$. This shows by induction that T_k is a polynomial of degree k for all $k \geq 0$. $\square$